

# VELMÈRE BASIC AUDIT REPORT

Ethereum Execution Protocol (Native Protocol Base Asset (Layer-1))

VELMERE SECURITY AUDIT REPORT: ETHEREUM EXECUTION PROTOCOL

Audited Contract: Native Protocol Base Asset (Layer-1)  
Network: Ethereum Mainnet (Chain ID: 1)  
Report ID: exec\_256\_eth\_basic\_real\_markets | Tier: BASIC | Application Surface: REAL MARKETS  
Created At: 2026-09-07T21:10:52.651Z

VERDICT SUMMARY: NOT SCORED (MISSING BYTECODE)  
Confidence Score: 0/100 | Evidence Coverage: 0%  
Bytecode is missing or malformed. In accordance with Velmère truth invariants, bytecode-derived claims are NOT emitted and security risk is marked NOT SCORED.

--- PROTOCOL OVERVIEW & NATIVE L1 NETWORK CONTEXT [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS  
Summary of distributed architecture and consensus mechanism  
Protocol / Asset: Ethereum Execution Protocol  
Architecture Type: Native Base Layer (Layer-1 · No Smart Contract)  
Network Identifier: native\_chain:eth  
Native Symbol: ETH  
Rule Governance: Distributed node consensus across validators / miners

--- CONSENSUS SPECIFICATION & NODE CLIENT VERIFICATION [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS  
Analysis of block validation rules, transactions, and cryptographic curves  
Verification of Ethereum Execution Protocol is conducted against native Layer-1 consensus specifications. EVM bytecode decompilation is NOT\_APPLICABLE for native base protocol assets.

--- BASELINE NETWORK SECURITY FINDINGS [BASIC] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS  
Identified risk vectors in base layer architecture

--- CONSENSUS GOVERNANCE & PROTOCOL UPGRADE ANALYSIS (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS  
Decentralization evaluation, validator distribution, and improvement proposal process  
[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- \* Distributed consensus governance and protocol upgrade mechanism analysis
  - \* Validator / miner distribution and network finality threshold evaluation
  - \* Hard-fork coordination history and client implementation diversity audit
  - \* WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
  - \* BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- SPOT MARKET DEPTH, EXCHANGE RESERVES & NETWORK VELOCITY (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS  
Global market liquidity, supply velocity, and mempool dynamics  
[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- \* Global order book depth across regulated spot venues and custodial reserves
- \* On-chain circulating supply velocity and active address distribution
- \* Network transaction fee volatility and mempool congestion behavior
- \* WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.

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--- NETWORK ATTACK VECTORS & CONSENSUS RESILIENCE (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Modeling of chain reorganizations, Sybil attacks, and P2P partitioning

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

## Summary of Analytical Depth in this section:

- \* Consensus reorganization and majority hashrate/stake attack modeling
  - \* P2P network partition, eclipse attack, and sybil resistance analysis
  - \* Mempool front-running, transaction censorship, and MEV extraction vectors
  - \* WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
  - \* BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).
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--- STATE ENGINE DETERMINISM & MULTI-CLIENT WIRE COMPATIBILITY (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Binary compatibility verification and state transition engine determinism

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

## Summary of Analytical Depth in this section:

- \* Reference node client reproducible build verification against repository tags
  - \* Consensus serialization and wire protocol binary compatibility analysis
  - \* State transition engine deterministic execution boundary checks
  - \* WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
  - \* BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).
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--- PROTOCOL HARD/SOFT FORK EVOLUTION & INVARIANT PROOFS (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Tracking of consensus rule evolution and backward compatibility guarantees

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

## Summary of Analytical Depth in this section:

- \* Protocol upgrade proposal activation and deployment history
  - \* Backward compatibility verification of block validation logic
  - \* Formal verification: NOT EXECUTED (Formal verification engine not commissioned)
  - \* WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
  - \* BLOCKER: TIER\_ENTITLEMENT\_BOUNDARY (client active tier does not cover this analytical layer).
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--- HUMAN ANALYST REVIEW & PROTOCOL ARCHITECTURE AUDIT (ADVANCED) [ADVANCED] ---

VERIFICATION TYPE: INDEPENDENT MANUAL AUDITOR REVIEW

Attestation by independent distributed systems auditor

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

## Summary of Analytical Depth in this section:

- \* Analyst sign-off requires verified reviewer evidence
  - \* Cryptographically signed analyst attestation digest
  - \* Manual review boundary explicit: no guarantee of absolute perfection
  - \* WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
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